

of pounds sterling, the Raspberry Pi set its goal at a fraction of the cost. In order to do this the computer system was stripped of all peripherals and accessories. The product only supplied the basic electronic hardware and the programming environment which would allow it to interface with other required devices.

Early Prototypes

The early prototype was based off an Atmel ATmega 644 microcontroller board in 2006. The next functional ARM version was in fact as small as the size of a pen drive with only two ports - one HDMI and one USB. The Atmel microcontroller was only clocked at 22.1 MHz with 512K SRAM for storage and buffer memory. The video signal capable on this set up was at a measly 320x240 resolution and could only generate simple graphics output. Within a couple of years, as mobile processors become cheaper and more affordable the project moved onto using ARM chips.

The first design on the Raspberry Pi was the Model B. In order to realise the goal for the Model B, Eben Upton approached RISC OS Open Community in 2011 to help develop the ports for the device. Eventually, Broadcom's Adrian Lees took on the challenge and even went on to contribute to the boards graphics capabilities. By August 2011, the first functional alpha boards of the Model B were in production but were larger in form factor due to hardware issues. Despite the size problem the boards were capable of running Quake 3 at Full HD(1920x1080) and processing Full HD video streaming over HDMI.

By October 2011, RISC OS 5 had been developed and was expected to be ported on to the Raspberry Pi within the year. Over the course of that year, multiple limited production runs were carried out for the Pi. The first Model B Beta boards came out of production in December 2011 but suffered from a single hardware error that was immediately rectified. Upon completion the Beta boards proved successful at playing Full HD video and measured up on the OpenGL ES benchmarking for graphics processing. As the final polish and fixes were implemented the launch data for the Pi drove ever closer. As a charitable and promotional step, 10 boards were put on auction in January 2012. The auction raised over GBP 16,000 on a retail value of GBP 220.

Just before the sales date the Raspberry Pi was fitted with the ability to run an operating system off an SD card in February 2012. The initial OS in play was the Debian 6.0 or Squeeze edition with the LXDE desktop, Midori